



use by 500 times and latency by 100 times compared to AI processors, while keeping main processors idle to extend battery life. With sub-milliwatt power, Pulsar supports gesture recognition, object detection, and presence detection, using 600 μ W for radar and 400 μ W for audio.

Innatera
<https://innatera.com>

Chipset combining RFID communication and AD conversion

RAMXEED has introduced an ASSP chipset that combines two functional chips—an RFID communication chip and an AD converter chip—on a compact 1.04mm² LSI with FeRAM. This integration enables wireless data transfer and direct sensor connectivity on a single platform, allowing for seamless integration. By providing a common architecture, the



chipset simplifies the development and manufacturing of smart contact lenses. Developers can combine it with various sensors to collect and process data directly within the lens, supporting a wide range of real-time monitoring and data-driven applications.

RAMXEED
<https://www.ramxeed.com>

Low-power memory for mobile AI applications

Micron Technology, Inc. has launched the first 1 γ (1-gamma) node-based low-power double data rate 5X (LPDDR5X) memory. It delivers 10.7Gbps speed, 20% power savings, and a package thickness of 0.61 millimeters—6% thinner than competitors and 14%

thinner than the previous generation. These features allow faster data transfer, lower energy use, and a smaller footprint.



Smartphone manufacturers can design thinner or foldable devices, while users benefit from longer battery life and smoother AI tasks, such as translation and image generation. In tests with Llama 2, the memory showed 30% faster location-based searches, over 50% faster voice-to-text translations, and up to 25% faster responses for purchase recommendations, enabling more efficient mobile AI applications.

Micron Technology, Inc.
<https://www.micron.com>

321-layer 1Tb 4D NAND flash for mobile devices

SK hynix has developed its UFS 4.1 solution using 321-layer 1Tb triple-level cell 4D NAND flash for mobile devices. The



product offers a sequential read speed of 4300MB/s, with 15% faster random read and 40% faster random write speeds. It has a thickness of 0.85mm and achieves 7% better power efficiency compared to the 238-layer version. The higher read and write speeds support data handling for AI workloads and multitasking. The thinner design allows integration into mobile devices. Better power efficiency reduces energy use during operation.

SK hynix
<https://skhynix.com>

3rd Gen automotive 5G module for satellite connectivity

LG Innotek has developed a third-generation automotive 5G communication

module that integrates satellite connectivity over 5G broadband. The module allows vehicles to communicate with terrestrial base stations and satellites



using broadband satellite communication (NR-NTN). It delivers transmission speeds in tens of megabits per second and reduces latency from seconds to hundreds of milliseconds. By combining terrestrial and satellite networks, the module ensures data exchange even when vehicles move beyond network coverage. Its transmission speeds are hundreds of times faster than those of previous satellite communication modules, while latency reduction enables faster data responsiveness.

LG Innotek
<https://www.lginnotek.com>

Cost-effective FPGA and SoC solutions

Microchip Technology has introduced PolarFire Core Field-Programmable Gate Arrays (FPGAs) and System-on-Chip (SoC) solutions. Built on the PolarFire



families, these devices remove serial transceivers and optimise features. By removing transceivers, the devices lower costs by up to 30% while maintaining performance. SEU immunity supports mission-critical reliability. The quad-core RISC-V MPU handles various compute workloads. Compatibility with PolarFire devices simplifies design reuse and reduces development effort. Customers can reduce costs while meeting performance, security, and reliability needs. The devices support control, edge processing, and safety functions across automotive, industrial automation, medical, communications, defence, and aerospace sectors.

Microchip Technology
<https://www.microchip.com>